

The Yearly Finance Equation

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Abstract

This paper introduces the Yearly Finance Equation. The paper ends with “The End”

Introduction

The **Yearly Finance Equation** is

$$N(1 - T) = S + I(1 + R) + E$$

where

$N \geq 0$ is Income

$0 \leq T \leq 1$ is Tax Rate

$S \geq 0$ is Savings

$I \geq 0$ is Investments

$R \geq 0$ is Return on Investments

$E \geq 0$ is Expenses

A Solution to the Yearly Finance Equation

Contrary to popular belief, there exists a solution to the Yearly Finance Equation.

$$N(\mu, \sigma, \tau) = \begin{cases} \frac{\exp(-\frac{(\mu - \ln \tau)^2}{2\sigma^2})}{\sqrt{2\pi}\sigma\tau} & \text{if } \tau > 0 \\ 0 & \text{if } \tau \leq 0 \end{cases} \quad (1)$$

$$T(\tau) = \begin{cases} \frac{2\tau \exp(-\frac{\tau^4}{\pi})}{\pi} & \text{if } \tau > 0 \\ 0 & \text{if } \tau \leq 0 \end{cases} \quad (2)$$

$$S(s, \tau) = \begin{cases} \frac{2s \exp(-\frac{s^2\tau^2}{\pi})}{\pi} & \text{if } \tau > 0 \\ 0 & \text{if } \tau \leq 0 \end{cases} \quad (3)$$

$$I(i, \tau) = \begin{cases} \frac{2i \exp(-\frac{i^2\tau^2}{\pi})}{\pi} & \text{if } \tau > 0 \\ 0 & \text{if } \tau \leq 0 \end{cases} \quad (4)$$

$$E(e, \tau) = \begin{cases} \frac{2e \exp(-\frac{e^2\tau^2}{\pi})}{\pi} & \text{if } \tau > 0 \\ 0 & \text{if } \tau \leq 0 \end{cases} \quad (5)$$

$$R(w, a, r, \tau) = \begin{cases} \frac{a}{w} \left(\frac{w}{\tau + w - r} \right)^{1+a} & \text{if } \tau > r \\ 0 & \text{if } \tau \leq r \end{cases} \quad (6)$$

Then $\exists \mu, \sigma, s, i, e, w, a, r, \tau :$

$$N(\mu, \sigma, \tau) (1 - T(\tau)) = S(s, \tau) + I(i, \tau) (1 + R(w, a, r, \tau)) + E(e, \tau)$$

The End